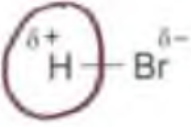
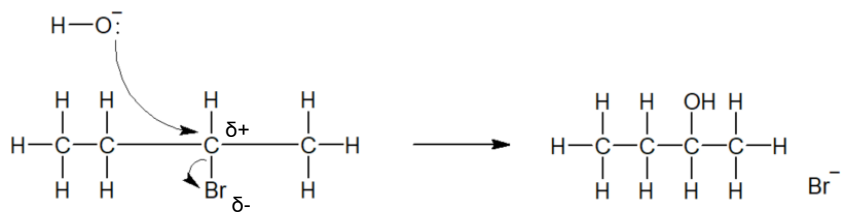


Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		 do not accept HBr fully circled	1			1		
		(ii)		heterolytic	1			1		
	(b)	(i)		 award (1) for curly arrows award (1) for partial charges		2		2		
		(ii)		aqueous sodium hydroxide (1) reflux / heating / boiling (1) must have 'hydroxide' as reagent to award this mark	2			2		2
		(iii)		secondary	1			1		
	(c)	(i)		oxidising (agent)	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		butan-2-ol has hydrogen bonding but butanone does not (1) therefore butanone has a lower <u>boiling</u> temperature (than butan-2-ol) (1) neutral answer – different boiling temperatures		2		2		
	(d)	(i)		product is lost at the end of each step (so two steps will mean a greater loss) accept references to percentage yield for both steps neutral answer – any reference to atom economy			1	1		
		(ii)	I	dehydration accept elimination do not accept condensation	1			1		
			II	$\begin{array}{c} \text{H} & & \text{CH}_2\text{CH}_3 \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{H} & & \text{H} \end{array}$ but-1-ene $\begin{array}{c} \text{CH}_3 & & \text{CH}_3 \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{H} & & \text{H} \end{array}$ (Z)-but-2-ene $\begin{array}{c} \text{CH}_3 & & \text{H} \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{H} & & \text{CH}_3 \end{array}$ (E)-but-2-ene award (3) for all structures and names correct award (2) for any four correct structures/names award (1) for any two correct structures/names do not accept correct name for an incorrect structure except in the case of methylpropene award maximum (2) if no mention of E/Z		3		3		
				Question 6 total	7	7	1	15	0	2